

UNIVERSIDAD
NACIONAL
DE COLOMBIA

Prototype 2

Advanced Architectural Structure

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Software Architecture

2026-II

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1. Objective

The objective of the second delivery of the project is to **redesign the architecture** and generate a second **prototype** of the software system developed as a project.

2. Requirements

2.1. Functional Requirements

- The functional completeness for the software system must be defined by the team.

2.2. Non-Functional Requirements

- The software system must follow a **distributed** architecture.
- The software system must include at least two different **presentation**-type **components** (one of them: web front-end).
- The web front-end must follow an SSR (Server-Side Rendering) **subarchitecture**.
- The software system must include a number of **logic**-type **components** at least equal to the number of team members.
- The software system must include at least one **component** that allows **communication/orchestration** between the logical components.
- The software system must include at least four **data**-type **components** (including relational and NoSQL databases).
- The software system must include at least one **component** that is responsible for handling **asynchronous** processes within the system.
- The software system must include a set of **HTTP**-based **connectors**.
- The software system must be built using at least five different general-purpose programming languages.

- The deployment of the software system must be **container**-oriented.

3. Delivery

3.1. Artifact

- Team
  - Name (1a, 1b, ..., 2a, 2b, ...)
  - Full names of the team members.
- Software System
  - Name
  - Logo
  - Description
- Architectural Structures
  - Component-and Connector (C&C) Structure
    - C&C View
    - Description of architectural elements and relations.
    - Description of architectural styles and patterns used.
  - Deployment Structure
    - Deployment View
    - Description of architectural elements and relations.
    - Description of architectural patterns used.
  - Layered Structure
    - Layered View
    - Description of architectural elements and relations.
    - Description of architectural patterns used.
  - Decomposition Structure
    - Decomposition View
    - Description of architectural elements and relations.
    - Description of architectural patterns used.
- Prototype
  - List of instructions (commands) to download the respective artifacts, and to run the software system.
  - Link to presentation video:
    - The video must last a minimum of **10 minutes** and a maximum of **15 minutes**.
    - The video must present the **architectural design** of the software system in this Prototype 2, based on the following architectural structures:
      - Component-and-Connector (C&C) Structure.
      - Deployment Structure.
      - Layered Structure.
      - Decomposition Structure.
    - The video must present the **software system in execution**, from:
      - The **functional** point of view (the main functionalities of the system).
      - The **deployed components** point of view (the running elements and their interactions).
    - The video must show the **consistency** between the presented architectural design and the built software system.

3.2. Submission Instructions

- The artifact must be created in **.md** format and exported to **.pdf** format.
- The artifact must be named **p2_X.pdf** (where X = the team name [1a, 1b, ..., 2a, 2b, ...]).
- The artifact must be submitted via MiCampus UNAL.
- Only one member of the team must submit the artifact.

3.3. Delivery Deadline

Tuesday, November 3, 2026, before 23h59.

3.4. Delivery Presentation

Tuesday, November 3, 2026 (video).